Web Checkout (KPG-2)

This documentation details the process of implementing the latest e-Payment Checkout platform released by Khalti.

How it works?

- User visits the merchant's website to make some purchase
- A unique `purchase_order_id` is generated at merchant's system
- Payment request is made to Khalti providing the `purchase_order_id`, `amount` in paisa and `return_url`
- User is redirected to the epayment portal (eg. https://pay.khalti.com)
- After payment is made by the user, a successful callback is made to the `return_url`
- The merchant website can optionally confirm the payment received
- The merchant website then proceeds other steps after payment confirmation

Getting Started

There is no special installation plugin or SDK required for this provided you are able to make a POST request from your web application. However, we shall come up with handy plugins in coming days.

> **Tip**
>
> A merchant account is required for integration.

Access Information

> **For Sandbox Access**

Signup from here as a merchant.

Please use 987654 as login OTP for sandbox env.

> **For Production Access**

Please visit here

Test Credentials for sandbox environment

> **Test Khalti ID for** 9800000000 9800000001 9800000002 9800000003 9800000004 9800000005
>
> **Test MPIN** 1111
>
> **Test OTP** 987654

Demo Flow for Checkout

Pay with Khalti

API Authorization

HTTP Authorization for api requests is done using Auth Keys. Auth Key must be passed in the header for authorization in the following format

```
{
    "Authorization": "Key <LIVE_SECRET_KEY>"
}
```

Tip

Use `live_secret_key` from **test-admin.khalti.com** during sandbox testing and use `live_secret_key` from **admin.khalti.com** for production environments.

API Endpoints

API Endpoints

Sandbox

https://dev.khalti.com/api/v2/

Production

https://khalti.com/api/v2/

Initiating a Payment request

Every payment request should be first initiated from the merchant as a server side `POST` request. Upon success, a unique request identifier is provided called `pidx` that should be used for any future references

URL	Method	Authorization	Format
/epayment/initiate/	POST	Required	application/json

JSON Payload Details

Field	Required	Description
return_url	Yes	• Landing page after the transaction. • Field must contain a URL.
website_url	Yes	• The URL of the website. • Field must contain a URL.
amount	Yes	• Total payable amount excluding the service charge. • Amount must be passed in Paisa
purchase_order_id	Yes	Unique identifier for the transaction generated by merchant
purchase_order_name	Yes	This is the name of the product.

Field	Required	Description
customer_info	No	This field represents to whom the txn is going to be billed to.
amount_breakdown	No	Any number of labels and amounts can be passed but the sum of amount_breakdown.amount mount be equal to amount.
product_details	No	No of set is unlimited

Sample Request Payload

```
{
  "return_url": "https://example.com/payment/",
  "website_url": "https://example.com/",
  "amount": 1300,
  "purchase_order_id": "test12",
  "purchase_order_name": "test",
  "customer_info": {
      "name": "Khalti Bahadur",
      "email": "example@gmail.com",
      "phone": "9800000123"
  },
  "amount_breakdown": [
      {
          "label": "Mark Price",
          "amount": 1000
      },
      {
          "label": "VAT",
          "amount": 300
      }
  ],
  "product_details": [
      {
          "identity": "1234567890",
          "name": "Khalti logo",
          "total_price": 1300,
          "quantity": 1,
   "unit_price": 1300
      }
  ],
  "merchant_username": "merchant_name",
  "merchant_extra": "merchant_extra"
}
```

Additionally Configuration also accepts attribute starting with `merchant_` that can be used to pass additional (meta) data.

- `merchant_name` : This is merchant name
- `merchant_extra` : This is extra data

The additional data starting with `merchant_` is returned in success response payload.

Examples

cURL

```
curl --location 'https://dev.khalti.com/api/v2/epayment/initiate/' \
--header 'Authorization: key 05bf95cc57244045b8df5fad06748dab' \
--header 'Content-Type: application/json' \
--data-raw '{
"return_url": "http://example.com/",
"website_url": "http://example.com/",
"amount": "1000",
"purchase_order_id": "Ordwer01",
"purchase_order_name": "Test",
"customer_info": {
    "name": "Test Bahadur",
    "email": "test@khalti.com",
    "phone": "9800000001"
}
}'
```

php

```
<?php
    $curl = curl_init();
    curl_setopt_array($curl, array(
    CURLOPT_URL => 'https://dev.khalti.com/api/v2/epayment/initiate/',
    CURLOPT_RETURNTRANSFER => true,
    CURLOPT_ENCODING => '',
    CURLOPT_MAXREDIRS => 10,
    CURLOPT_TIMEOUT => 0,
    CURLOPT_FOLLOWLOCATION => true,
    CURLOPT_HTTP_VERSION => CURL_HTTP_VERSION_1_1,
    CURLOPT_CUSTOMREQUEST => 'POST',
    CURLOPT_POSTFIELDS =>'{
    "return_url": "http://example.com/",
    "website_url": "https://example.com/",
    "amount": "1000",
    "purchase_order_id": "Order01",
        "purchase_order_name": "test",

    "customer_info": {
        "name": "Test Bahadur",
        "email": "test@khalti.com",
        "phone": "9800000001"
    }
```

```
    }

    ',
    CURLOPT_HTTPHEADER => array(
        'Authorization: key live_secret_key_68791341fdd94846a146f0457ff7b455',
        'Content-Type: application/json',
    ),
    ));

    $response = curl_exec($curl);

    curl_close($curl);
    echo $response;
```

python

```
import requests
import json

url = "https://dev.khalti.com/api/v2/epayment/initiate/"

payload = json.dumps({
    "return_url": "http://example.com/",
    "website_url": "https://example.com/",
    "amount": "1000",
    "purchase_order_id": "Order01",
    "purchase_order_name": "test",
    "customer_info": {
    "name": "Ram Bahadur",
    "email": "test@khalti.com",
    "phone": "9800000001"
    }
})
headers = {
    'Authorization': 'key live_secret_key_68791341fdd94846a146f0457ff7b455',
    'Content-Type': 'application/json',
}

response = requests.request("POST", url, headers=headers, data=payload)

print(response.text)
```

C#

```
using System;
using System.Net.Http;
using System.Text;
using System.Threading.Tasks;
using Newtonsoft.Json;
namespace KhaltiApiExample
{
    class Program
    {
        static async Task Main(string[] args)
        {
            var url = "https://dev.khalti.com/api/v2/epayment/initiate/";
```

```
            var payload = new
            {
                return_url = "http://example.com/",
                website_url = "https://example.com/",
                amount = "1000",
                purchase_order_id = "Order01",
                purchase_order_name = "test",
                customer_info = new
                {
                    name = "Ram Bahadur",
                    email = "test@khalti.com",
                    phone = "9800000001"
                }
            };

            var jsonPayload = JsonConvert.SerializeObject(payload);
            var content = new StringContent(jsonPayload, Encoding.UTF8,
"application/json");

            var client = new HttpClient();
            client.DefaultRequestHeaders.Add("Authorization", "key
live_secret_key_68791341fdd94846a146f0457ff7b455");

            var response = await client.PostAsync(url, content);
            var responseContent = await response.Content.ReadAsStringAsync();

            Console.WriteLine(responseContent);
        }
    }
}
```

NodeJs

```
var request = require('request');
var options = {
    'method': 'POST',
    'url': 'https://dev.khalti.com/api/v2/epayment/initiate/',
    'headers': {
    'Authorization': 'key live_secret_key_68791341fdd94846a146f0457ff7b455',
    'Content-Type': 'application/json',
    },
    body: JSON.stringify({
    "return_url": "http://example.com/",
    "website_url": "https://example.com/",
    "amount": "1000",
    "purchase_order_id": "Order01",
    "purchase_order_name": "test",
    "customer_info": {
        "name": "Ram Bahadur",
        "email": "test@khalti.com",
        "phone": "9800000001"
    }
    })

};
request(options, function (error, response) {
```

```
    if (error) throw new Error(error);
    console.log(response.body);
});
```

Success Response

```
    {
        "pidx": "bZQLD9wRVWo4CdESSfuSsB",
        "payment_url": "https://test-pay.khalti.com/?
pidx=bZQLD9wRVWo4CdESSfuSsB",
        "expires_at": "2023-05-25T16:26:16.471649+05:45",
        "expires_in": 1800
    }
```

After getting the success response, the user should be redirected to the `payment_url` obtained in the success response.

Error Responses

return_url is blank

```
{
    "return_url": [
        "This field may not be blank."
    ],
    "error_key": "validation_error"
}
```

return_url is invalid

```
{
    "return_url": [
        "Enter a valid URL."
    ],
    "error_key": "validation_error"
}
```

website_url is blank

```
{
    "website_url": [
        "This field may not be blank."
    ],
    "error_key": "validation_error"
}
```

website_url is invalid

```
{
    "website_url": [
        "Enter a valid URL."
    ],
    "error_key": "validation_error"
}
```

Amount is less than 10

```
{
    "amount": [
        "Amount should be greater than Rs. 10, that is 1000 paisa."
    ],
    "error_key": "validation_error"
}
```

Amount is invalid

```
{
    "amount": [
        "A valid integer is required."
    ],
    "error_key": "validation_error"
}
```

purchase_order_id is blank

```
{
    "purchase_order_id": [
        "This field may not be blank."
    ],
    "error_key": "validation_error"
}
```

purchase_order_name is blank

```
{
    "purchase_order_name": [
        "This field may not be blank."
    ],
    "error_key": "validation_error"
}
```

Amount breakdown doesn't total to the amount passed

```
{
    "amount": [
        "Amount Breakdown mismatch."
    ],
    "error_key": "validation_error"
}
```

Payment Success Callback

After the user completes the payment, the success response is obtained in the return URL specified during payment initiate. Sample of success response return URL.

- The callback url `return_url` should support `GET` method
- User shall be redirected to the `return_url` with following parameters for confirmation
  - pidx - *The initial payment identifier*
  - status - *Status of the transaction*
    - Completed - Transaction is success
    - Pending - Transaction is in pending state, request for lookup API.
    - User canceled - Transaction has been canceled by user.
  - transaction_id - *The transaction identifier at Khalti*

  - tidx - *Same value as transaction id*
  - amount - *Amount paid in paisa*
  - mobile - *Payer KHALTI ID*
  - purchase_order_id - *The initial purchase_order_id provided during payment initiate*
  - purchase_order_name - *The initial purchase_order_name provided during payment initiate*
  - total_amount - *Same value as amount*
- There is no further step required to complete the payment, however merchant can process with their own validation and confirmation steps if required
- It's recommended that during implementation, payment lookup API is checked for confirmation after the redirect callback is received

Sample Callback Request

- Success transaction callback

```
http://example.com/?pidx=bZQLD9wRVWo4CdESSfuSsB
&txnId=4H7AhoXDJWg5WjrcPT9ixW
&amount=1000
&total_amount=1000
&status=Completed
&mobile=98XXXXX904
&tidx=4H7AhoXDJWg5WjrcPT9ixW
&purchase_order_id=test12
&purchase_order_name=test
&transaction_id=4H7AhoXDJWg5WjrcPT9ixW
```

- Canceled transaction callback

```
http://example.com/?pidx=bZQLD9wRVWo4CdESSfuSsB
&transaction_id=
&tidx=
&amount=1000
&total_amount=1000
&mobile=
&status=User canceled
&purchase_order_id=test12
&purchase_order_name=test
```

Important

- Please use the lookup API for the final validation of the transaction.
- Khalti payment link expires in 60 minutes in production (default).

Payment Verification (Lookup)

After a callback is received, You can use the `pidx` provided earlier, to lookup and reassure the payment status.

URL	Method	Authorization	Format
/epayment/lookup/	POST	Required	application/json

Request Data

```
{
   "pidx": "HT6o6PEZRWFJ5ygavzHWd5"
}
```

Success Response

```
{
   "pidx": "HT6o6PEZRWFJ5ygavzHWd5",
   "total_amount": 1000,
   "status": "Completed",
   "transaction_id": "GFq9PFS7b2iYvL8Lir9oXe",
   "fee": 0,
   "refunded": false
}
```

Pending Response

```
{
   "pidx": "HT6o6PEZRWFJ5ygavzHWd5",
   "total_amount": 1000,
   "status": "Pending",
   "transaction_id": null,
   "fee": 0,
   "refunded": false
}
```

Initiated Response

```
{
   "pidx": "HT6o6PEZRWFJ5ygavzHWd5",
   "total_amount": 1000,
   "status": "Initiated",
   "transaction_id": null,
   "fee": 0,
   "refunded": false
}
```

Refunded Response

```
{
   "pidx": "HT6o6PEZRWFJ5ygavzHWd5",
   "total_amount": 1000,
   "status": "Refunded",
   "transaction_id": "GFq9PFS7b2iYvL8Lir9oXe",
   "fee": 0,
   "refunded": true
}
```

Expired Response

```
{
   "pidx": "H889Er9gq4j92oCrePrDwf",
   "total_amount": 1000,
   "status": "Expired",
   "transaction_id": null,
   "fee": 0,
   "refunded": false
}
```

Canceled Response

```
{
   "pidx": "vNTeXkSEaEXK2J4i7cQU6e",
   "total_amount": 1000,
   "status": "User canceled",
   "transaction_id": null,
   "fee": 0,
   "refunded": false
}
```

Payment Status Code

Status	Status Code
Completed	200
Pending	200
Expired	400
Initiated	200
Refunded	200

Status	Status Code
User canceled	400
Partially Refunded	200

Lookup Payload Details

Status	Description
pidx	This is the payment id of the transaction.
total_amount	This is the total amount of the transaction
status	`Completed` - Transaction is success `Pending` - Transaction is failed or is in pending state `Refunded` - Transaction has been refunded `Expired` - This payment link has expired `User canceled` - Transaction has been canceled by the user `Partially refunded` - Transaction has been partially refunded by the user
transaction_id	This is the transaction id for the transaction. This is the unique identifier.
fee	The fee that has been set for the merchant.
refunded	`True` - The transaction has been refunded. `False` - The transaction has not been refunded.

Lookup status

Field	Description
Completed	Provide service to user.
Pending	Hold, do not provide service. And contact Khalti team.

Field	Description
Refunded	Transaction has been refunded to user. Do not provide service.
Expired	User have not made the payment, Do not provide the service.
User canceled	User have canceled the payment, Do not provide the service.

Important

- Only the status with **Completed** must be treated as success.
- Status **Canceled** , **Expired** , **Failed** must be treated as failed.
- If any negative consequences occur due to incomplete API integration or providing service without checking lookup status, Khalti won't be accountable for any such losses.
- For status other than these, hold the transaction and contact **KHALTI** team.
- Payment link expires in 60 minutes in production.

Generic Errors

When an incorrect Authorization key is passed.

```
{
   "detail": "Invalid token.",
   "status_code": 401
}
```

If incorrect pidx is passed.

```
{
   "detail": "Not found.",
   "error_key": "validation_error"
}
```

If key is not passed as prefix in Authorization key

```
{
    "detail": "Authentication credentials were not provided.",
```

```
    "status_code": 401
}
```